ILAPARTI RAJASEKHAR

Portfolio

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CAREER OBJECTIVE:

Motivated and enthusiastic third-year B.Tech student in Electronics and Communication Engineering with hands-on experience building embedded systems and IoT devices using ESP32, Arduino, PIC, and Raspberry Pi Pico. Seeking internship opportunities to learn, grow, and contribute to real-world embedded and IoT products.

EDUCATIONAL QUALIFICATIONS

S.No	Name of the course	Name of the Institution	Year of passing	Percentage
1	B.Tech(ECE)	Seshadri Rao Gudlavalleru Engineering college	2028	7.31 CGPA
2	Intermediate	Sri Chaitanya Jr college	2024	93.2%
3	SSC	AANM & VVRSR GUDLAVALLERU[EM] High school	2022	90.33%

TECHNICAL SKILLS

- **Programming Languages**: C, Embedded C, C++, Python, Verilog
- **Embedded Systems**: PIC16F877A, Arduino UNO / Nano, ESP32 / ESP32-C3, ESP8266, Raspberry Pi Pico
- **Sensors & Interfaces**: I2C, UART, MPU6500, ACS712, HC-SR04, MQ-2, DHT11, LCD / OLED, relays, servos
- **Tools & Software**: MPLAB IDE, Arduino IDE, Proteus, Vivado, Icarus Verilog, GTKWave, Git / GitHub, MS Office, Canva, Photoshop, Lightroom
- **AI Tools**: Claude / Claude Code, ChatGPT, GitHub Copilot, Google Gemini

INTERNSHIP EXPERIENCE

Embedded Systems Intern — Emertxe Information Technologies

- Developed a PIC16F877A-based Microwave Oven Controller featuring keypad input, LCD display, and timer-driven alerts.
- Implemented state-machine logic for cooking modes and safety protocols.
- Debugged firmware via MPLAB IDE and validated hardware through simulation and deployment.

PROJECTS

- **Electrical Pole Safety Alert System** (ESP32-C3, MPU6500, ACS712) Detects pole tilt, impact, free-fall, and current hazards with buzzer alerts, NVS event history, and a Wi-Fi dashboard.

- 🔗 **AgriGate – Automated Water Gateway** (Arduino Nano, ESP32) Water-level-based servo gate control for paddy fields with automatic drainage, runoff detection, OLED status, and a web dashboard.
- 🔗 **Plant Care Rover** (Raspberry Pi Pico, ESP32-C3) Web-controlled rover with autonomous mode, servo soil sampling, and automatic watering; boards communicate over UART.
- 🔗 **FireGuard UNO** (Arduino Uno) Fire and gas monitor using MQ-2, flame, and DHT11 sensors that automatically drives a fan, pump, and buzzer through relays.
- 🔗 **Water Treatment Turbidity Monitor** (ESP32-C3) Turbidity station with three-point calibration, alarms, NVS data logging, LCD readout, and a web dashboard with trends.
- 🔗 **Smart Nursery Watering System** (ESP32-C3) Soil-moisture-based automatic irrigation with hysteresis, pump safety cutoff, LittleFS watering history, and a Wi-Fi dashboard.
- 🔗 **Distance Alert System** (ESP32-C3, HC-SR04) Ultrasonic distance monitor with adaptive buzzer/LED alerts and a browser-adjustable threshold.
- 🔗 **LAN File Share** (Python, Windows app) Shares a PC folder with phones on the same Wi-Fi via QR code, with file previews, ZIP downloads, phone uploads, and PIN protection.
- 🔗 **OLED Video Maker** (Web app, Web Serial) Browser tool that converts videos into 128×64 OLED animations and flashes them to ESP32 / Arduino boards over USB.
- 🔗 **Hush** (Web app) Private messaging web app with user accounts, deployed on Netlify.

ADDITIONAL EXPERIENCE

Freelance Photo Editor — Fiverr

- Delivered international photo-editing projects with consistent **5-star client ratings**.
- Managed client communication, deadlines, and quality assurance.

ACHIEVEMENTS & CERTIFICATIONS

- Internship Completion – Emertxe Information Technologies
- Hackathon Participant & Paper Presenter
- Photography Certification – MindLuster

STRENGTHS

Problem Solving • Team Collaboration • Technical Presentation • Debugging • Time Management • Creative Eye

INTERESTS

Embedded Systems • IoT • Automation • Photography • Tech Innovation

Personal details

DOB	:	31st August 2006
Nationality	:	Indian
Gender	:	Male
Marital status	:	Single
Languages Known	:	Telugu, English
Hobbies	:	Photography, Traveling, Cooking, Reading Novels

Declaration : I hereby declare that the above information is true to the best of my Knowledge

Place : Gudlavalleru

Date : [ILAPARTI RAJASEKHAR]